Corrigé — Seven-Day Discrete Deployment Gates

Do not use this file before a complete written attempt. The canonical TD corrigé remains the source for TD 03 Exercises 0–8; this document solves only the packet-specific Entry Gate and Exercises A–E. After locating the first rupture, close the solution and reconstruct the decisive step without notes on the following day.

Entry Gate

1. Define

$$p(x) = \mu(\{x\}), \qquad x \in E.$$

   Countable additivity on the disjoint singleton decomposition gives

$$\mu(A) = \sum_{x \in A} p(x) = \int_A p \, d\#_E.$$

   Thus $\mu = p\#_E$. The density is unique $\#_E$-almost everywhere, hence pointwise because the only $\#_E$-null subset of a countable space is empty.

2. On $[0,1]$, $\#(A) = 0$ implies $A = \varnothing$, so $\lambda \ll \#$. If $\lambda = f\#$, testing the singleton $\{x\}$ gives

$$f(x) = \int_{\{x\}} f \, d\# = \lambda(\{x\}) = 0$$

   for every x. Hence $f = 0$, contradicting $\lambda([0,1]) = 1$. The unavailable RN hypothesis is σ-finiteness of the reference counting measure.

3. For the law $\mu_X = X_\# \mathbb{P}$,

$$\mathbb{E}[g(X)] = \int g \, d\mu_X,$$

   or $\sum_x g(x) p_X(x)$ in the atomic representation. In general this is not $g(\mathbb{E}[X])$.

4. For integer-valued (X, Y),

$$p_{X+Y}(n) = \sum_{k \in \mathbb{Z}} p_{X,Y}(k, n-k).$$

   If X and Y are independent, $p_{X,Y} = p_X p_Y$, and only then does the expression reduce to $p_X * p_Y$.

5. The constructions give, respectively: binomial; hypergeometric; negative binomial in the trial convention; a mixed Poisson law. Under the stated square-integrability hypothesis, a non-degenerate intensity contributes the strictly positive between-component term $\mathrm{Var}(\Lambda)$, so the mixture cannot be Poisson.

Exercise A — Identical marginals, different pushforwards

Both coordinates of both pairs equal either B or $1 - B$, so all four marginals are Bernoulli$(1/2)$.

For the positively coupled pair,

$$p_{X_+,Y_+}(0,0) = p_{X_+,Y_+}(1,1) = \frac{1}{2},$$

and all other joint masses vanish. Therefore

$$\mathbb{P}(X_+ + Y_+ = 0) = \mathbb{P}(X_+ + Y_+ = 2) = \frac{1}{2}.$$

For the negatively coupled pair,

$$p_{X_-,Y_-}(0,1) = p_{X_-,Y_-}(1,0) = \frac{1}{2},$$

so $X_- + Y_- = 1$ almost surely. Neither pair is independent: for example,

$$p_{X_+,Y_+}(0,0) = \frac{1}{2} \neq \frac{1}{4}, \qquad p_{X_-,Y_-}(0,1) = \frac{1}{2} \neq \frac{1}{4}.$$

For $h(x,y)=\mathbf{1}_{\{x+y\geq 1\}}$, one has

$$h(X_+,Y_+)=B, \qquad h(X_-,Y_-)=1.$$

Thus the first pushforward is Bernoulli$(1/2)$ with expectation $1/2$, whereas the second is δ_1 with expectation 1. In both cases the source measure is the joint law on $\{0,1\}^2$, not the pair of marginal laws.

More generally, every joint law with Bernoulli$(1/2)$ marginals has, for some $a\in[0,1/2]$,

$$p_{00}=p_{11}=a, \qquad p_{01}=p_{10}=\frac{1}{2}-a.$$

Hence

$$\mathbb{P}(X+Y=0)=a, \quad \mathbb{P}(X+Y=1)=1-2a, \quad \mathbb{P}(X+Y=2)=a.$$

The marginals determine the support and the mean $\mathbb{E}[X+Y]=1$, but not the sum law.

Exercise B — Finite-population correction

The support is $\{0,1,2,3,4,5\}$ and

$$\mathbb{P}(H=k)=\frac{\binom{7}{k}\binom{13}{5-k}}{\binom{20}{5}}.$$

For ordered-draw success indicators $I_1,\ldots,I_5$,

$$H=\sum_{j=1}^{5} I_j, \qquad \mathbb{E}[I_j]=\frac{7}{20},$$

so

$$\mathbb{E}[H]=5\frac{7}{20}=\frac{7}{4}.$$

For $i\neq j$,

$$\operatorname{Cov}(I_i,I_j)=-\frac{7}{20}\frac{13}{20}\frac{1}{19}=-\frac{91}{7600}.$$

Therefore

$$\operatorname{Var}(H)=5\frac{7}{20}\frac{13}{20}\frac{15}{19}=\frac{273}{304}.$$

A binomial$(5,7/20)$ variable has variance

$$5\frac{7}{20}\frac{13}{20}=\frac{91}{80}.$$

The difference is

$$\frac{91}{80}-\frac{273}{304}=\frac{91}{380},$$

which is exactly the negative of

$$2\sum_{1\leq i<j\leq 5}\operatorname{Cov}(I_i,I_j)=-\frac{91}{380}.$$

If $n=N$, then $H=K$ almost surely and the finite-population factor $N-n$ makes the variance zero. If $N=1$, every admissible model is deterministic; one uses that construction directly rather than the quotient containing $N-1$.

Exercise C — Finite and infinite waiting structures

To have the second success at position t, one success must occur among the first $t-1$ positions, position t must be a success, and the remaining three successes must occur among the last $12-t$ positions. Thus

$$\mathbb{P}(T_2^{(12)} = t) = \frac{\binom{t-1}{1}\binom{12-t}{3}}{\binom{12}{5}},$$

with exact support

$$2 \le t \le 9.$$

In the iid Bernoulli(5/12) model,

$$\mathbb{P}(T_2 = t) = \binom{t-1}{1}\left(\frac{5}{12}\right)^2\left(\frac{7}{12}\right)^{t-2}, \qquad t \ge 2.$$

The two laws differ because the finite model has bounded support and because its draws are negatively dependent with exactly five total successes; the iid model has an infinite horizon and a fixed success probability at every trial.

For $n \ge 2$,

$$T_2 \le n$$

holds exactly when the first n trials contain at least two successes. Hence

$$\{T_2 \le n\} = \{\mathrm{Bin}(n, 5/12) \ge 2\}.$$

The failures-before-second-success variable is $F_2 = T_2 - 2$ and has support $\mathbb{N}$. At $p = 0$, the second success time is $+\infty$ almost surely and no finite-valued negative-binomial law results. At $p = 1$, $T_2 = 2$ and $F_2 = 0$ almost surely.

Exercise D — One Poisson model through three representations

For $k, l \in \mathbb{N}$,

$$\begin{aligned}
\mathbb{P}(K = k, L = l) &= \mathbb{P}(N = k + l)\binom{k+l}{k}\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^l \\
&= e^{-4}\frac{(4/3)^k}{k!}\frac{(8/3)^l}{l!} \\
&= \left(e^{-4/3}\frac{(4/3)^k}{k!}\right)\left(e^{-8/3}\frac{(8/3)^l}{l!}\right).
\end{aligned}$$

Therefore

$$K \sim \mathrm{Pois}(4/3), \qquad L \sim \mathrm{Pois}(8/3),$$

and the factorization proves independence.

Conditioning on $K + L = 5$ and cancelling the common Poisson factors gives

$$K \mid (K + L = 5) \sim \mathrm{Bin}(5, 1/3).$$

Since

$$\frac{K(K+1)}{2} = \frac{(K)_2}{2} + K,$$

and $\theta = 4/3$,

$$\mathbb{E}[Y] = \frac{\theta^2}{2} + \theta = \frac{20}{9}.$$

On the other hand,

$$g(\mathbb{E}[K]) = g(4/3) = \frac{14}{9}.$$

The missing amount is

$$\frac{2}{3} = \frac{1}{2}\operatorname{Var}(K).$$

If labels cease to be conditionally independent, the binomial allocation given N, the joint factorization, the Poisson marginals and their independence can fail. The identity $K + L = N$ survives, and the first moments survive if each event still has the stated marginal label probability independently of N.

Exercise E — Mixture and truncation

For $k \in \mathbb{N}$,

$$\mathbb{P}(N = k) = \frac{3}{4}e^{-1}\frac{1}{k!} + \frac{1}{4}e^{-4}\frac{4^k}{k!}.$$

Let $\Lambda = 1$ when $I = 0$ and $\Lambda = 4$ when $I = 1$. Then

$$\mathbb{E}[\Lambda] = \frac{7}{4}, \qquad \operatorname{Var}(\Lambda) = \frac{1}{4}\frac{3}{4}(4-1)^2 = \frac{27}{16}.$$

Consequently,

$$\mathbb{E}[N] = \frac{7}{4}, \qquad \operatorname{Var}(N) = \mathbb{E}[\Lambda] + \operatorname{Var}(\Lambda) = \frac{55}{16}.$$

Since $55/16 \neq 7/4$, the unconditional law cannot be Poisson.
The two joint masses on the observed fibre $\{N = 0\}$ are

$$\mathbb{P}(I = 0, N = 0) = \frac{3}{4}e^{-1}, \qquad \mathbb{P}(I = 1, N = 0) = \frac{1}{4}e^{-4}.$$

Normalization gives

$$\mathbb{P}(I = 1 \mid N = 0) = \frac{e^{-4}}{3e^{-1} + e^{-4}} = \frac{1}{3e^3 + 1}.$$

For $M \sim \operatorname{Pois}(\lambda)$ with $\lambda > 0$, set $a = 1 - e^{-\lambda}$. The zero-truncated pmf is

$$\mathbb{P}(M = k \mid M \geq 1) = \frac{e^{-\lambda}\lambda^k}{a\,k!}, \qquad k \geq 1.$$

Because the zero term contributes nothing to the first two raw moments,

$$\mathbb{E}[M \mid M \geq 1] = \frac{\lambda}{a},$$

and

$$\begin{aligned}\operatorname{Var}(M \mid M \geq 1) &= \frac{\lambda^2 + \lambda}{a} - \frac{\lambda^2}{a^2} \\ &= \frac{\lambda}{1 - e^{-\lambda}} - \frac{\lambda^2 e^{-\lambda}}{(1 - e^{-\lambda})^2}.\end{aligned}$$

The latent mixture is a convex sum of two probability measures. Truncation is restriction of one measure to $\{1, 2, \ldots\}$ followed by normalization. Their numerators retain Poisson factors for different structural reasons; neither operation produces a new member of the original Poisson family in the non-degenerate regime.

Official-audit structural key

This table classifies the mechanism only. It does not reproduce or answer the official questions.

Question	Structural object
Q30	fixed independent Bernoulli trials; binomial construction
Q128	sampling without replacement; hypergeometric count
Q146	finite-horizon event expressed through a negative-binomial waiting structure and conditioning
Q150	discrete-uniform law followed by conditioning or finite mixing
Q227	Poisson mixture and total variance
Q246	independent Poisson components conditioned on their total; binomial allocation
Q386	nonlinear Poisson payment evaluated through factorial moments
Q512	second moment distinguished from the square of the mean

Use the official sample solutions only to verify the locked numerical answers.

Canonical TD solution map

Canonical exercise	Decisive reconstruction
TD 03 Exercise 0	uncountable counting measure is not σ-finite; $\lambda \ll \#$ has no density
TD 03 Exercise 1	normalized atomic weights define a countably additive measure and its integral
TD 03 Exercise 2	fibre pushforward and the injectivity boundary
TD 03 Exercise 3	addition pushforward; convolution only under the product joint law
TD 03 Exercise 4	binomial versus hypergeometric construction and covariance correction
TD 03 Exercise 5	infinite waiting law versus finite-population order statistic
TD 03 Exercise 6	Poisson factorial moments, superposition, splitting and conditional allocation
TD 03 Exercise 7	mixture overdispersion, Bayes reversal and zero truncation
TD 03 Exercise 8	latent class, conditional thinning, mixture and nonlinear payment

For the complete proofs, use the corresponding section of Corrigé TD 03 after the written attempt. This packet does not maintain a second copy.